

SECTION 15951
PIPING TESTING, FLUSHING, CLEANING, PARTICLE TESTING

PART 1 – GENERAL

1.1 SUMMARY

- A. Section Includes:
 - 1. Testing, Flushing, and Cleaning requirements for the following systems:
 - a. Chilled Water (with and without Glycol),
 - b. Condenser Water (with and without Glycol),
 - c. Hot Glycol,
 - d. Tempered Water, Non-Potable Water, Potable Water
 - e. High Resistivity Water
 - f. Process Vacuum
 - g. Compressed Dry Air, High Pressure Compressed Dry Air, Instrument Air
 - h. Natural Gas
 - i. Solvent Waste
 - j. Fire Protection
 - k. Waste Drains for IW, HFW, DIRC, and NPW
 - 2. Particle Testing:
 - a. Compressed Dry Air, High Pressure Compressed Dry Air
- B. Systems not covered:
 - 1. UPW and HUPW: Refer to Section 15091.
 - 2. Waste Collection/Treatment Systems in Polypropylene Piping: Refer to Section 400513.77.
 - 3. PFA Systems: Refer to Section 15092.
 - 4. Fire Protection: Refer to Section 15300.
 - 5. High Purity Gas Systems: Refer to Section 15372.
 - 6. General Purity Gas Systems: Refer to Section 15373.
- C. Related Documents:
 - 1. Section 15077 - Labeling Requirements for Systems and Equipment.
 - 2. Section 15090 - PVC and CPVC Piping Systems.
 - 3. Section 15091 - PVDF Piping Systems.
 - 4. Section 15092 - PFA Plastic Piping Systems.
 - 5. Section 15094 - Pipe Supports and Hangers.
 - 6. Section 15101 - Carbon Steel Piping Systems.
 - 7. Section 15300 - Fire Protection Systems.
 - 8. Section 15372 - High Purity Stainless Steel Tubing Systems.
 - 9. Section 15373 - General Purity Stainless Steel Tubing Systems.
 - 10. Section 400513.76- FRP Piping Systems.

1.2 DEFINITIONS

- A. Design Pressure:
 - 1. The design pressure for each component in a piping system is equal to the system pressure at the most severe condition.
 - 2. Piping not protected by pressure relieving devices, and/or that can be isolated from a pressure-relieving device, shall be designed for the highest pressure that can be developed.
 - 3. Sources of pressure variation to consider include ambient influences, pressure oscillations and surges, improper operation, and failure of control devices.
- B. Design Temperature: The design temperature for each component in a piping system is equal to the system temperature at the most severe condition.

1.3 SUBMITTALS

- A. Submit the following in accordance with Division 1:
 - 1. Submit written Pipe Testing, Flushing, and Cleaning Procedures prior to performing test.
 - 2. Submit sign-off for system cleaning certified by cleaning contractor.
 - 3. Submit written Pipe Particle Testing Procedures.
 - 4. Submit design for high point vents/low point drains on double contained systems.
- B. Submit the following Informational Submittals after completion of tests:
 - 1. Written results of testing specified as part Field Quality Control articles.
 - 2. Certified test reports of cleaning and testing for each system.
 - 3. Certification of potable water system disinfection.

1.4 QUALITY ASSURANCE

- A. Designate a quality control representative (QCR) to inspect the installation and document cleaning and testing of piping systems. QCR shall be qualified as welding inspector per AWS-QC2, or ASNT Level-2 certified. Maintain on site records for CM or Owner review.
 - 1. Test piping systems that are governed by ASME B31.1, B31.3, or B31.9 in accordance with the applicable code.
 - 2. Test Natural Gas piping in accordance with ANSI Z223.1/NFPA 54, National Fuel Gas Code, IPC, IMC, and all local utility company requirements.
 - 3. Test fire protection systems in accordance with applicable NFPA codes.
 - 4. Test potable water, sanitary waste, storm, and vent piping per IBC and this section.
 - 5. Follow the most stringent requirements where testing requirements differ.
- B. Except where noted in Specific System Requirements, conduct pneumatic tests for 24-hour duration without pressure degradation and hydrostatic tests for 4-hour duration without pressure degradation. Perform pneumatic tests at 1.1 times design pressure and hydrostatic tests at 1.5 times design pressure.
- C. Pipelines which tie into existing systems and which cannot be pressure tested shall be tested to the existing system tie-in point. The tie-in connection when welded shall be 100% radiograph inspected in accordance with AWS.
- D. For short runs, as approved by the CM or Owner Representative, a 100% X-ray examination may be substituted for a pressure test.

- E. Comply with referenced codes and manufacturer's standards (Most applicable based on adoption by AHJ).
 - 1. ANSI/ASME B31.1, Power Piping.
 - 2. ANSI/ASME B31.3, Process Piping.
 - 3. ANSI/ASME B31.9, Building Services Piping.
 - 4. International Building Code (IBC).
 - 5. International Mechanical Code (IMC).
 - 6. International Plumbing Code (IPC).

PART 2 – PRODUCTS

2.1 MATERIALS

- A. The Subcontractor shall furnish all materials, instruments, and equipment (except government furnished material and equipment) required to perform the pressure tests. All test equipment shall have been calibrated within 30 days of use, be in good working order and have gauges accurate to within + 3.0% of span: All test gauges shall have a mirror-band 4.5 in dial with knife-edge pointer, thus eliminating parallax visual error, with 1-psi increments.
- B. The Subcontractor is responsible for all pipe taps, bypasses, pancakes, drains, vents, and any other materials or piping modifications required for the testing, flushing, cleaning, and particle analysis for the systems covered in this section.
 - 1. Provide vents and drains at all high and low points in each system required to fill and drain for testing, cleaning, flushing, and operation.
 - 2. Vents and drains for pipes 2 inches and smaller shall be 1/2 inch, for pipes 2-1/2 inches and larger shall be 3/4 inch. Provide ball valve, hose connection, and cap on drains and vents.

PART 3 – EXECUTION

3.1 GENERAL

- A. Scheduling:
 - 1. Do not cover or enclose the piping work before it has been cleaned, tested, inspected, and approved by the CM.
 - 2. Testing shall be performed after fabrication, prior to insulation or other finish work and before equipment is initially placed in service. All pressure joints (including welds) except those noted otherwise shall be exposed during tests. If approved by Owner, installing contractor may choose to insulate prior to testing; regardless, all welds and mechanical connects shall be exposed during testing.
 - 3. Perform tests in the presence of CM and Owners' Rep. Document procedures and test results for record. Submit in accordance with Para. 1.3.
 - 4. Testing spool pieces is allowed, but does not replace a complete system test.
 - 5. There are only two acceptable joints (equipment connection and the source connection) that can be excluded from the complete system test.
 - 6. Any components that cannot be tested through (check valve, regulator, etc.) must be replaced with a spool for final system test.
 - 7. Snoop leak detection fluid cannot be used in the fab or on any high purity system. Use a combination of Triton X-100 surfactant as a substitute for non-high purity systems.
 - 8. Triton X-100 mix, Snoop, or any other liquid detection media must not be used on high

purity systems!

9. All future valves/ports must be capped or plugged.
10. Testing components and material must consist of equal to or higher quality than the engineered specs for the system being tested.
11. All testing equipment must be installed in such a way that it does not interfere with or run the risk of coming in contact with existing components.
12. Prior to pressure testing, thoroughly examine completed pipe system for the following:
 - a. System is complete as per drawings with correct components, O-rings, and supports in place
 - b. Adequate pipe spacing per Section 15094.
 - c. Labeling is complete per Section 15077.

B. Safety:

1. When conditions require that a pressure test be maintained for a period of time during which the testing medium in the system would be subject to thermal expansion, provisions shall be made for the relief of the excess pressure so caused.
2. If pressure vessels are rated for conditions equal to or exceeding the maximum operating conditions of the connecting piping system, conduct tests including the vessel(s); otherwise, isolate the vessels from the piping system for testing.
3. Do not use butterfly valves as blanking devices for pressure testing. Blind-flanges, or inline pancakes are approved blanking devices.
4. Recognize the hazards associated with pneumatic pressure testing and take necessary precautions to protect all personnel and adjacent systems in the event of a joint failure.
5. Open all vents when draining piping to avoid pulling a vacuum on the piping.
6. Overnight hydrostatic testing is NOT allowed.
7. Prior to testing, safety precautions must be instituted to protect personnel and property in case of test failure.
8. Do not use compressed air or other gases to test any systems consisting of PVC, CPVC, PVDF, PP, or PE.
9. Important: Make sure high point vents are always installed in wet systems.
10. A JHA must be filled out prior to hydrostatic testing. During the JHA review, look for the following:
 - a. Electrical sources (support equipment, electrical panels, etc.)
 - b. Personnel occupied space (anything under eight feet from a walking surface requires signage)
11. A job plan must be in place to establish stations where attendants are needed while filling the system to verify that water has displaced the air at all high points (high points, drain valves).
12. The piping system must be adequately supported to limit movement. The system may require additional anchoring at a change of direction.
13. The piping system must be filled with water, taking care to prevent surge and air entrapment.
14. All trapped air must be slowly released. Vents must be provided at all high points of the piping system.
15. All valves and air-relief mechanisms shall be opened so that air can be vented while the system is being filled.
16. Trapped air is extremely dangerous and it must be slowly and completely vented prior to testing.
17. High point vents and low point drains shall be designed into the piping system by engineering. However, all vents and/or drains may not be shown on the drawings and it's up to the installing contractor to provide vents or drains as needed, and/or directed by the CM or owners' rep.
18. Care must be taken to ensure the pressure does not exceed the working pressure of the lowest rated component in the system (valves, unions, flanges, etc.).
19. Any leaking joints or pipe must be cut out and replaced. No patchwork! If there is a

leak, replace the joint and start the testing process from the beginning.

C. Gauges:

1. Building Mechanical and Plumbing piping shall be tested at pressures and durations listed within each section below.
2. Maintain calibration records for gauges utilized for pressure testing.
3. Conduct tests with 2 gauges installed as close as possible to the low point of the piping system.

1.3 SPECIFIC SYSTEM REQUIREMENTS

A. Chilled Water Systems (with or without Glycol), Condenser Water Systems (with or without Glycol), Hot Glycol Systems, Tempered Water, Non-Potable Water, Potable Water.

1. Install flushing pump suction and discharge spool pieces to piping for flushing (pump must be supplied to support needed flow in step 12).
2. Install temporary bypass loops around all chillers, towers, heat exchangers, boilers, and miscellaneous equipment to ensure adequate flushing velocities (if applicable).
3. Note these bypasses or spools are not shown on construction drawings but are required and are the responsibility of the construction contractor.
4. Under no circumstances shall water be flushed through chillers, cooling towers, heat exchangers, boilers, or any equipment that will act as a trap for debris.
5. Insert temporary cone strainers upstream of equipment or components not directly protected by system strainers such as pumps (if applicable).
6. Air test piping: Gradually increase air pressure until 25 psig is reached.
7. Verify and repair any leaks prior to next step.
8. Hydrostatic test piping: Fill piping with Non-potable water (Potable water for PW system) while venting all air from all high point vents.
9. Bring up to test pressure shown in Table 1. Monitor for time shown. Contact Owner Representative for filling. Contractor shall not turn valves connected to active systems.
10. Drain water to an acceptable location through drain valves. Coordinate with CM.
11. Refill with cleaning chemicals: Use no solvents in system cleaning, use only normally accepted and environmentally safe and acceptable chemicals that could be introduced into any waste system. Use degreasing and de-oiling agents in conjunction with passivating chemicals.
12. Circulate cleaning chemicals for at least twelve hours at not less than 3 ft./s up to 18 inches diameter, or 2 ft./s for 20 inches diameter and larger. Provide circulation pump.
13. Note while re-circulating chemicals the water temperature needs to be monitored to make sure overheating doesn't occur.
14. Drain chemical water and refill with clean water.
15. Circulate clean water for at least 2 hours at not less than 3 ft./s for pipe up to 18 inches diameter and at not less than 2 ft./s for 20 inches diameter and larger.
16. Drain water and take sample to Owner Representative or CM for inspection: If water is acceptable to Owner Representative or CM, move on to step 17, if water is unacceptable repeat steps 14 through 16 until water is acceptable.
17. Remove temporary strainers and clean/reinstall permanent strainers.
18. Remove temporary bypass loops around equipment installed in step 2. Install permanent piping to this equipment.
19. Clean and flush all dead legs, valves and flanges that were not able to be flushed during steps (1-16) above.
20. Fill with final water or glycol/water mix.
21. Hydrostatic test again as per step 9 for final acceptance.

B. Condensate Return (steam condensate):

1. Disconnect condensate piping from receiver tanks, deaerators, or any equipment that will act as a trap for debris (if applicable). Install blind flanges or pancakes for testing.
2. Air test piping: Gradually increase air pressure until 25 psig is reached.

3. Repair any leaks prior to proceeding.
4. Hydrostatic test piping: Fill piping with Non-potable water while venting all air from all high point vents. Bring up to test pressure shown in Table 1. Monitor for time shown. Contact Owner Representative for filling. Contractor shall not turn valves tied to active systems.
5. Drain water piping to an approved location. Coordinate with CM.
6. Connect a steam line to the condensate piping and add vents per the following. Note these connections may not be on the construction drawings but are required and are the responsibility of the construction contractor.
 - a. Condensate piping sizes 2" through 6": Connect to steam via 1" steam pipe with 2" vent to outside on the far end. Install valves on all connections.
 - b. Condensate piping sizes 8" and larger: Connect to steam via 2" steam pipe with 4" vent to outside on the far end. Install valves on all connections.
7. Flush condensate piping using steam: Open steam and vent valves and flow steam for 8 hours or until steam on far end is clean. Coordinate with CM.
8. Disconnect steam from condensate piping. If no equipment (tanks, deaerators, etc.) is being installed in the project then testing/flushing is complete. If equipment is being installed in project, move onto next step.
9. Make all final connections to equipment.
10. Re-Hydrostatic test piping per step "c" through equipment.
11. Drain water to an approved location. Coordinate with CM.

C. Steam System:

1. Disconnect steam piping from boilers (if applicable). Install blind flanges or pancakes for testing.
2. Air test piping to 25 psig: Gradually increase air pressure until a pressure of 25 psig is reached. Walk entire system to verify that no leaks are present and repair any leaks found.
3. Air test piping to final pressure:
 - a. Gradually increase air pressure until the test pressure shown in Table 1.
 - b. Monitor for time shown.
 - c. Contractor shall not turn valves tied to active systems.
4. Relieve pressure from system.
5. Make supply and vent connections per the following.
 - a. Note these connections may not be on the construction drawings but are required and are the responsibility of the construction contractor.
 - b. Steam piping sizes 2" through 6": Install 1-1/2" steam connection and 3" vent to outside on far end. Install valves on all connections.
 - c. Steam piping sizes 8" and larger: Install 3" steam connection and 6" vent to outside on far end. Install valves on all connections.
6. Flush steam line using steam: Open steam and vent valves and flow steam for 8 hours or until steam out far end is clean. Coordinate with CM.
7. Remove steam flushing line. If no equipment (boilers, etc.) is being installed in the project then testing/flushing is complete. If equipment is being installed then move onto the next step.
8. Make final connections to equipment (remove blind flanges or pancakes installed in step 1).
9. Follow steps 2 thru 4 again.

D. High Resistivity System (metal piping):

1. Install pump suction and discharge spool pieces to piping for flushing.
2. Install pancakes and temporary bypass loops around heat exchangers to ensure adequate flushing velocities (if applicable). Note these bypasses or spools are not shown on construction drawings but are required and are the responsibility of the construction contractor. Under no circumstances shall water be flushed through heat

- exchangers or any equipment that will act as a trap for debris that cannot be removed.
3. Insert temporary cone strainers upstream of equipment or components not directly protected by system strainers such as pumps (if applicable).
4. Air test piping: Gradually increase air pressure until 25 psig is reached.
5. Verify and repair any leaks prior to next step.
6. Hydrostatic test piping: Fill piping with DI/RO water while venting all air from all high point vents. Bring up to test pressure shown in Table 1. Monitor for time shown.
7. Contact Operations Coordinator (66400) for filling.
8. Contractor shall not turn valves tied to active systems.
9. Drain water to acceptable location through drain valves.
10. Coordinate with Operations Coordinator (66400).
11. Refill with DI/RO water.
12. Circulate water for at least 12 hours at not less than 3 feet per second on pipe sizes 18 inches and smaller and at not less than 2 feet per second on pipe sizes 20 inches and larger.
13. Drain water and take sample to Owner Representative or CM for inspection: If water is acceptable to Operations Coordinator or CM, move on to next step, if water is unacceptable repeat steps 11 through 13 until water is acceptable. Note that water only needs to be circulated for 2 hours each time after initial 12-hour flushing.
14. Remove temporary strainers and clean/reinstall permanent strainers.
15. Remove temporary bypass loops around equipment installed in step 2.
16. Install permanent piping to this equipment. Keep pancakes on heat exchangers.
17. Fill with final DI/RO water.
18. Hydrostatic test again as per step 6 for final acceptance.
19. If hydrostatic test passes remove pancakes from heat exchangers.
20. Bring system to 10 psig and perform visual inspection for leaks at heat exchangers.

E. High Resistivity System (plastic piping):

1. Vacuum test piping: Install Vacuum pump and pull vacuum to (-23"Hg).
2. Monitor for 1 hour and fix any leaks prior to next step.
3. Hydrostatic test piping: Fill piping with DI/RO water while venting all air from all high point vents.
4. Bring up to test pressure shown in Table 1. Monitor for time shown in same table.
5. Contact Operations Coordinator (66400) for filling.
6. Contractors shall not turn valves tied to active systems.
7. Drain water to an acceptable location through drain valves. Coordinate with Owner and CM.
8. Refill with DI/RO water.
9. Drain water to acceptable location.
10. Refill with DI/RO water.
11. For branch lines 1" and smaller that are less than 20' vacuum test to (-23"Hg) and test at system pressure for a 4 hour duration is all that is required.
12. If a vacuum test shows a leak and the leak could not be identified, you must get a variance through "HAZDEMO" and engineering to perform a 5 psig air test.

F. Compressed Dry Air, High Pressure Compressed Dry Air, and Instrument Air Systems:

1. Air/Nitrogen test piping: Gradually increase air/nitrogen pressure until a pressure of 25 psig is reached. Walk entire system to verify that no leaks are present and repair any leaks found.
2. Air/Nitrogen test piping to final pressure: Gradually increase air/nitrogen pressure until the test pressure shown in table below is met. Monitor for time shown in same table.
3. After piping has passed air/nitrogen test, open end of line valves and flush all lines for 2 hours.
4. Note these lines are not to be open to the cleanroom.
5. Flush lines to create turbulent flow so to scrub the piping. See following chart for

flushing flow rates.

Turbulent Flow Requirements

Pipe Diameter, (inch)	Flow Rates (scfh)
0.5	75
0.75	116
1	162
1.5	255
2	349
3	535
4	715
6	1078

7. Test all 1" and larger CDA and HPCDA lines for particles and VOC emissions to the following:
 - a. 10 particles/ft³ maximum count at 0.2 micron and larger cumulative in a 1-minute period.
 - b. Must pass 5 consecutive tests.
 - c. VOC emissions: Not to exceed 3 ppb at 70 degrees F.
- G. Process Vacuum System:
 1. Install temporary vacuum pump to pull vacuum to pressure shown in Table 1.
 2. Shut valve and turn off pump.
 3. Monitor time at pressure shown.
- H. Natural Gas:
 1. Insert cone strainers upstream of equipment or components not protected by system strainers.
 2. Air test piping to 25 psig: Gradually increase air pressure until a pressure of 25 psig is reached. Walk entire system to verify that no leaks are present and repair any leaks found.
 3. Air test piping to final pressure: Gradually increase air pressure until the test pressure shown in Table 1. Monitor for time shown.
 4. Blow down pipes free from foreign substances before installing valves, stops, or making final connections. When blowing down (flushing) piping systems, use 100- psig clean dry air (CDA) (oil and moisture free) for natural gas piping system(s) and components.
 5. Clean all piping and equipment before installation.
 6. Remove any piece of equipment (filters, drip traps, burners, regulators, heat exchangers, etc.), which may get damaged as a result of this quantity of flow.
 7. Install loops to enable flushing or blow down activities as necessary.
 8. Remove and clean strainers.
 9. Repeat process until strainers show no signs of collecting additional debris.
- I. Abatement Lines:
 1. All process FORELINES and pumped exhaust piping are stainless steel.
 2. No pressure loss shall occur during testing in order for a line to be qualified.
 3. After the pressure test is conducted, an orange "tested tag" is to be attached to the exhaust/foreline.
 - a. For abatement lines that are insulated, place tags and labels in a Ziploc bag and tie wrap the bag to the KF25 port on the pump fore line spool piece.
 - b. After insulation is complete, apply labels and tags to the lines. It is important the line documentation remains with the line at all times.
 4. Plastic tags are installed stating the equipment number and port number
 - a. Port tag locations are inlet ports to abatement equipment, floor penetrations,

- ceilings, and wall penetrations.
- b. Tested tags for exhaust lines will be located at the point of use abatement equipment.
- c. If the exhaust line travels through a wall or floor penetration, it is required to place a "tested tag" at the point of use abatement equipment and at the back of the pump.
- d. Tag placement for forelines will be near the pump.

J. Aqueous Waste Drains:

1. Pumped drains:
 - a. Vacuum test piping: Install Vacuum pump and pull vacuum to (-23"Hg).
 - b. Monitor for 1 hour and fix any leaks prior to next step.
 - c. Hydrostatic test piping: Fill piping with non-potable water while venting all air from all high point vents. Contact Owner Representative for filling. Contractor shall not turn valves tied to active systems.
 - d. Drain to approved location. Coordinate with CM.
2. Gravity drains:
 - a. Vacuum test piping: Install Vacuum pump and pull vacuum to (-23"Hg).
 - b. Monitor for 1 hour and fix any leaks prior to next step.
 - c. Hydrostatic test piping: Fill piping with non-potable water while venting all air from all high point vents. Contact the Operations Coordinator at (66400) for filling.
 - d. Contractor shall not turn valves tied to active systems.
 - e. Drain to approved location. Coordinate with CM.

K. Solvent Waste Drains:

1. Pumped drain:
 - a. Air test to 25 psig: Gradually increase air pressure to 25 psig. Walk system to verify that no leaks are present and repair as necessary.
 - b. Gradually increase air to final test pressure shown in Table 1.
 - c. Monitor for time shown.
2. Gravity drain:
 - a. Air test to 25 psig: Gradually increase air pressure to 25 psig. Walk system to verify that no leaks are present and repair as necessary.
 - b. Gradually increase air to final test pressure shown in Table 1.
 - c. Monitor for time shown.

3.4 LEAK REPAIR

- A. Those welds, joints, surfaces, areas, etc. found to be unacceptable from the testing shall be repaired. Notify Contractor of any test failures.
 1. Repair leaks in soldered systems by melting out the joint, thoroughly cleaning parts, and/or replacing the fitting and re-soldering the joint.
 2. Repair minor leaks in welded joints by chipping out and re-welding.
 3. Repair leaks in threaded joints by replacing the thread, the fitting, or both.
- B. On flanged joints replace defective or damaged gaskets during testing and flushing.
 1. Replace all over-torqued plastic flanges.
 2. Use new gaskets each time a flanged joint is made up.
- C. Retest repaired piping systems until acceptance is obtained.
- D. Repair leaks which develop during the warrantee.

3.5 POTABLE WATER SYSTEMS STERILIZATION

- A. In addition to initial cleaning in 3.3.B.1 the following is required:
 - 1. Utilize a chlorine-water solution having a free chlorine residual of 50 ppm.
 - 2. Prepare mixture using calcium or sodium hypochlorite and water.
 - 3. Operate all valves, hydrants, and other appurtenances during sterilization to ensure that the mixture is dispersed into all parts of the system.
 - 4. Retain water in pipeline a minimum of 24 hours or long enough to destroy all non- spore forming bacteria. Residual chlorine at end of retention period shall be not less than 10 ppm.
 - 5. Flush sterilizing solution from the system utilizing the permanent source of supply until the system is chemically and bacteriologically equal to the source water.
 - 6. Provide adequate warning signs at points of possible use during system disinfection and flushing.
 - 7. Submit written certification of disinfecting completion.

3.6 ACCEPTANCE

- A. Maintain test records for each system tested, flushed, chemical treated, cleaned, or sterilized.
 - 1. Identify procedures, repairs, and test results.
 - 2. Obtain signature and date for all personnel performing and witnessing tests.
 - 3. Owner Representative to witness all testing.
- B. Acceptance of each system is contingent upon sign off by CM and Owner Representative for all required tests.

TABLE 1 – SYSTEM TESTING PARAMETERS

Pipe System	Pressure Pre-Test	Medium	Duration	Pressure	Medium	Duration
CHWS/R, PCHWS/R, TCHWS/R, CWS/R, GHWS/R, TW, NPW, Metal piping	25 psig	Air		150 psig	NPW	4 hr.
PW	25 psig	Air		150 psig	PW	4 hr.
CR (Condensate)	25 psig	Air		150 psig	NPW	4 hr.
MPS (Steam)	25 psig	Air		150 psig	NPW	24 hr.
HR (metal)	25 psig	Air		150 psig	DI/RO water	4 hr.
HR / PCW (PVC)	-23" Hg Vacuum	Vacuum	1 hr.	150 psig	DI/RO water	4 hr.
HR / PCW Branch (PVC), Size up to 1" and less than 20'	-23" Hg Vacuum	Vacuum	1 hr.	System > 60 psig	HR / PCW	4 hr.
PV	N/A	N/A	N/A	-23" Hg Vacuum	Vacuum	24 hr.
PV Branch	N/A	N/A	N/A	-23" Hg Vacuum	Vacuum	4 hr.
IA	25 psig	Air		150 psig	Air	4 hr.
BA	N/A	N/A		100 psig	CDA (no substitute)	4 hr.
CDA	25 psig	Air		150 psig	Air	4 hr.
CDA Branch with components	N/A	Air		100 psig	Air	4 hr.
HPCDA	25 psig	CDA/Nitrogen		180 psig	Nitrogen	4 hr.
NG	25 psig	Air		100 psig	Air	4 hr.
SW (pumped)	25 psig	Air		150 psig	Air	24 hr.
SW (gravity)	25 psig	Air		100 psig	Air	24 hr.
Waste Drains rated for 150 PSI PVC/CPVC (pumped IW, DIRC, HFW, NPW)	-23" Hg Vacuum	Vacuum		100 psig	NPW	4 hr.
Waste Drains for PVC/CPVC (gravity IW, DIRC, HFW, NPW)	-23" Hg Vacuum	Vacuum		30 psig	NPW	4 hr.
Dry Pump process Forelines and pumped exhaust Spool	N/A	N/A		25 psig	Air	0.5 hr.
Complete Pumped Exhaust	N/A	N/A		25 psig	Air	4 hr.
Complete Process Foreline gross leak test	N/A	N/A		25 psig	Air	1 hr.
Gas Equipment Vent Lines	N/A	N/A		100 psig	Nitrogen	4 hr.

END OF SECTION